

Semantic Bottlenecks in Visual EEG-to-Language Reconstruction

Anonymous submission

Abstract

We study whether visual EEG embeddings that support image reconstruction also support language reconstruction, and introduce a THINGS-EEG2 semantic bottleneck audit with real-text derangement nulls, declared candidate banks, and prototype retrieval before any text is generated. Direct supervision with detailed image captions fails: brain-derived embeddings obtain only 2.33 BLEU and 0.149 token-set F1 on a sub-08 diagnostic, below a 1,000-derangement real-text null by 2.40 BLEU and 0.073 F1. The failure is consistent with target-geometry mismatch: generated captions average 14.66 words and have only 0.582 within-class GTR cosine across images of the same object class, while class-level prototypes are short and stable. We therefore evaluate *Semantic Prototype Stabilization* (SPS) as a leakage-controlled protocol rather than an unrestricted sentence decoder: decode a normalized semantic vector, retrieve from a declared memory, and verbalize only after retrieval. Across ten subjects, SPS reaches $26.25 \pm 5.29\%$ top-5 in a closed 200-way diagnostic whose candidate bank contains the held-out target; in the more realistic no-test-target setting, a reproduced full-schedule MindAlign visual decoder with a validation-selected low-rank GTR bridge reaches 0.479 class-memory cosine and 0.292 raw-caption F1. A scaled real ZuCo sentence-window extension exposes an alignment pitfall: head-truncated sentence alignment fails, while tail alignment gives a weak rank signal, with ridge reaching 0.120 top-5 against a 0.064 real-text null and a 0.111 time-shifted EEG null. The bounded conclusion is that the current bottleneck is not GTR-to-text inversion but mismatch between neural evidence, target granularity, and alignment; defensible brain-to-text pipelines should expose semantic rank before open-ended generation.

Introduction

The long-term goal of brain-language decoding is to reconstruct semantically faithful language from neural activity. For non-invasive EEG, this goal is especially difficult: EEG has high temporal resolution but low spatial resolution, low signal-to-noise ratio, and strong subject variability. A practical route is therefore to avoid direct token decoding and instead map EEG into a foundation-model embedding space,

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then use a language model or embedding inversion model to verbalize the decoded representation.

Visual EEG reconstruction appears to make this route plausible. Recent EEG-to-image systems align neural signals with vision-language embeddings and use diffusion models to reconstruct perceived visual stimuli (Li et al. 2024a). This progress creates an evaluation risk: plausible output can be supplied by a strong generator prior even when the EEG evidence is weak. In the local THINGS-EEG pipeline, image reconstruction performance is strong enough to ask whether the same EEG can be mapped into a language embedding and decoded into a sentence. This paper does not claim open-ended sentence reconstruction from visual EEG; it asks which part of the modular route fails and which constrained semantic evidence remains.

We focus on a concrete pipeline: EEG is encoded by an ATMS-style neural encoder, optionally refined by a diffusion prior, mapped into a 768-dimensional GTR sentence embedding, and inverted into text with a pretrained Vec2Text-style corrector (Morris et al. 2023; Ni et al. 2021). The language targets are produced in two ways. The first uses a vision-language captioner to describe every stimulus image, then encodes the caption with GTR. The second uses the THINGS folder label, formatted as a short class-level sentence. The first target is visually rich and instance-specific; the second is coarse and concept-level.

The empirical result is a negative finding with a falsifiable diagnosis. Ground-truth GTR embeddings are easy to invert into the reference text, so text inversion is not the main failure. However, EEG-derived embeddings trained against detailed captions produce text below a real-text permutation null. Coarse labels expose concept-level structure but do not solve reconstruction: their fixed wording inflates text-overlap metrics even when the object label is wrong. This means the visual EEG signal carries some decodable category-level semantics, but the current evidence does not support the fine visual attributes and relations inserted by generated captions. In other words, direct EEG-to-caption embedding regression asks the brain signal to predict details that the visual EEG representation, at least in this setting, does not stably encode.

Our contributions are fourfold. First, we introduce a visual EEG-to-language semantic evidence audit with 1,000-derangement real-text nulls, ground-truth ceilings, direct-encoder controls, and explicit candidate-bank declarations.

Second, we quantify target-granularity mismatch as a measurable failure mode for generated-caption supervision. Third, we formalize *Semantic Prototype Stabilization* as a leakage-controlled protocol rather than an unrestricted sentence decoder: learn stable prototype semantics, retrieve from a declared memory, and only then verbalize. The held-out test-200 memory is used only as a closed rank diagnostic, while the training memory contains no held-out target labels. Fourth, we reproduce a full MindAlign visual decoding schedule and show that its EEG embeddings can be transferred into GTR semantic memories with training data only, while low-rank calibration and negative adapter, reranker, verbalizer, and real sentence-window route-decision audits identify which post-hoc repairs are bounded and which route must be rejected before sentence-generation claims are made.

Background and Motivation

Large visual EEG datasets make it possible to study neural responses to many object concepts and natural images (Grootswagers et al. 2022). THINGS-style visual EEG is well suited for object perception and concept-level decoding, and EEG-to-image reconstruction systems exploit this by aligning EEG to CLIP-like visual embeddings before generation (Li et al. 2024a). These embeddings contain high-level visual information, but they are not the same as sentence semantics.

Language embeddings such as GTR are trained for semantic retrieval: sentences with related meanings should have nearby vectors (Ni et al. 2021). Vec2Text further shows that text embeddings can be inverted into fluent text when the target embedding is valid and close to the training distribution (Morris et al. 2023). These properties suggest a modular brain-to-text pipeline: decode a language embedding from EEG, then invert the embedding into text. The key assumption is that the EEG signal contains enough information to identify the target language embedding. Our experiments test this assumption directly.

Relation to Brain-to-Text Decoding

Brain-to-text work spans different signal modalities and task paradigms, so direct metric comparisons can be misleading. fMRI semantic decoders reconstruct the meaning of continuous perceived or imagined speech by fitting subject-specific cortical semantic models and using language-model search (Tang et al. 2023). EEG-to-text work usually uses reading or language-comprehension datasets such as ZuCo, where the stimulus is already a sentence and the target text is time-aligned with EEG (Wang and Ji 2022; Wang et al. 2024; Zhou et al. 2024; Tao et al. 2024). Recent EEG and MEG language decoders also use LLMs, audio-text representations, or Whisper-style intermediate spaces to generate text from language-evoked brain signals (Li et al. 2024b). These works address language-to-brain-to-language reconstruction; our task asks whether image-evoked EEG can support text targets imposed after stimulus presentation.

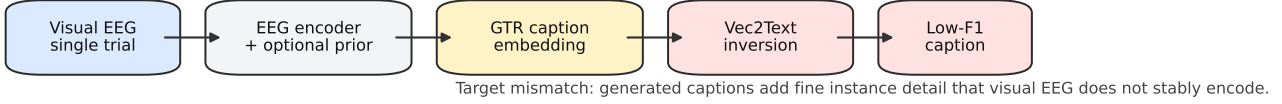
Several recent systems move in the same direction as our diagnosis: they reduce reliance on unconstrained autoregressive generation and add retrieval, semantic summarization, or intermediate representations. DELTA uses a vector-

quantized EEG tokenizer with a diffusion language model on ZuCo (Jeon and Kim 2025); GLIM explicitly reframes EEG-to-text as semantic summarization rather than verbatim reconstruction and evaluates EEG-text retrieval controls (Liu, Shen, and Liu 2025); SENSE maps EEG to sparse textual cues before prompting an LLM (Murhekar et al. 2026); Brain-CLIPLM decodes compressed semantic anchors before retrieval-grounded LLM reconstruction (Yang et al. 2026); BrainMosaic models semantic intent as a set of compositional units before sentence reconstruction (Li et al. 2026); and a 2026 RAG-based system retrieves sentence embeddings before LLM refinement on silent-reading EEG (Collautti et al. 2026). These methods support the broader principle that non-invasive brain-to-text should expose a grounded semantic bottleneck before generation. However, they are designed for language-evoked EEG or intent-decoding settings, where the text supervision is closer to the neural task than post-hoc captions for visual objects.

Our setting is different. The stimulus is an image, not a sentence, and the text target is imposed afterward by a captioner or class label. The closest visual EEG-to-text line is Thought2Text, which trains a ChannelNet encoder against CLIP image embeddings and object labels, then adapts instruction-tuned LLMs to generate captions from EEG embeddings on the six-subject CVPR2017 visual EEG dataset (Mishra et al. 2025). Its reported BLEU-1 scores of roughly 22.7–26.0% demonstrate that visual EEG plus strong visual-language priors can generate plausible descriptions, but the system also uses object-label and CLIP-aligned visual grounding. More recent visual decoding work such as MindAlign further shows that jointly aligning EEG, images, and generated text can improve THINGS-EEG2 visual retrieval (Chen et al. 2026); our exact-code ten-subject 150-epoch bridge reaches 81.78% visual top-5 and passes the predefined visual reproduction gate, but its text-feature top-5 is 28.05%, and neither metric is a generated-sentence score. Using the explicit subject-tagged checkpoints from this rerun, we map MindAlign EEG embeddings into GTR space with training data only and retrieve from no-test-target semantic memories; validation-selected calibration reaches 0.478 class-memory cosine and 0.291 raw-caption F1. This makes the key question not whether a visual-language prior can verbalize a visually grounded embedding, but whether the EEG representation itself preserves language-semantic neighborhoods. Recent critiques of EEG-to-text evaluation show that teacher forcing, weak baselines, and noise controls can make language generation appear stronger than the EEG evidence supports (Jo et al. 2024). We therefore use real-text derangement nulls, ground-truth embedding ceilings, direct-encoder controls, explicit candidate-bank declarations, and test-200 prototype retrieval to separate text-generator ability from brain-to-semantic evidence. This is a protocol-level comparison rather than a cross-dataset leaderboard: our claim is not public SOTA sentence generation, but a leakage-controlled diagnosis of why post-hoc captions are poor visual-EEG text targets.

We therefore treat SOTA positioning as a protocol audit rather than a leaderboard. The fair comparisons in this paper are same-split controls on THINGS-EEG2: real-text

A. Failure mode: direct caption embedding inversion



B. Semantic Prototype Stabilization: retrieve evidence before verbalization

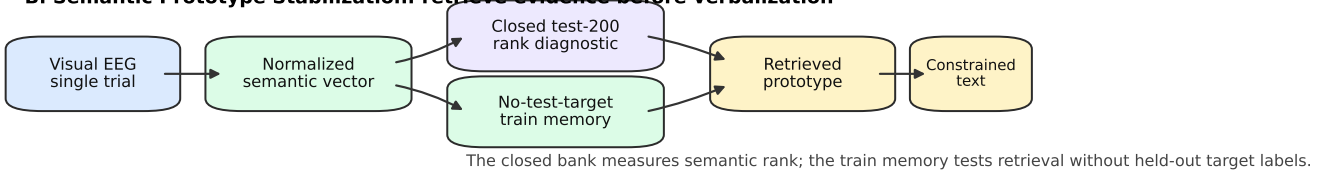


Figure 1: Protocol view of the THINGS-EEG2 visual EEG-to-language bottleneck. Direct inversion treats a noisy EEG-derived vector as if it were a valid detailed caption embedding. SPS instead separates evidence from verbalization: closed test-200 retrieval measures semantic rank when the held-out target is in the bank and is not a deployable reconstruction setting, while no-test-target training-memory retrieval tests constrained verbalization without held-out labels as candidates.

derangement nulls, GTR ceilings, diffusion prior versus direct encoder, and test-200 semantic prototype retrieval under identical subject scope.

Benchmark

Data and Signal

We use the local THINGS-EEG2-style preprocessing used by the EEG image reconstruction pipeline. The visual task contains EEG responses to natural object images, with 1,654 training object classes, 16,540 training images, and a held-out test set of 200 disjoint object classes. Following the existing generation pipeline, EEG epochs are passed to an ATMS encoder and evaluated on subject `sub-08` with seeds 42, 123, and 456 when available. We also report a matched 80-epoch semantic-prototype sweep on `sub-01`–`sub-10` to test whether the route runs across all available subjects under an identical short training schedule.

Language Targets

We evaluate three target regimes. *Detailed captions* are generated per image by a vision-language model and encoded with `sentence-transformers/gtr-t5-base`. *Caption class mean* replaces each training image’s caption embedding by the mean embedding over the 10 images in its object class. *Class labels* encode the short folder-label sentence “This picture is [class]”. These regimes form a granularity ladder from instance-level visual description to concept-level semantics.

Model and Controls

The training path is EEG to ATMS encoder to diffusion prior to GTR embedding. The predicted embedding is decoded by a pretrained GTR Vec2Text corrector. We evaluate four controls. The *brain* suite uses predicted EEG embeddings. The Gaussian suite inverts random Gaussian embeddings and is retained only as an inversion stress test. The main textual chance model is a *real-text derangement null*: each reference is compared with a mismatched held-out caption

or label from the same target regime. The *ceiling* suite inverts the ground-truth GTR embedding of the target text. A high ceiling with low brain performance means the text inversion module is functional and the bottleneck lies in EEG-to-language alignment, but the derangement null is the appropriate baseline for caption-level text metrics.

We also evaluate a *direct encoder* suite before the diffusion prior to test whether an image-reconstruction prior preserves or degrades language-space neighborhoods. The main ten-subject SPS sweep uses seed 42, 80 training epochs, early-stopping patience 20, GTR-T5-base text targets, and the same train/test image split for every subject. Generation-mode training uses an MSE-heavy objective with loss weights exposed in the run script; retrieval-mode training L2-normalizes EEG and text embeddings and uses multi-positive contrastive targets so repeated class prototypes are not treated as false negatives. All model-selection diagnostics use training-validation artifacts before test evaluation.

The retrieval-style objective is

$$\mathcal{L} = \lambda \|z - g(y)\|_2^2 + (1 - \lambda) \mathcal{L}_{\text{MP-InfoNCE}}(\bar{z}, \overline{g(y)}),$$

where z is the EEG-derived embedding, bars denote L2 normalization, and the multi-positive InfoNCE term treats all samples sharing the same prototype text as positives. The generation setting uses the local MSE-heavy blend because Vec2Text is sensitive to absolute embedding position, while retrieval-mode experiments emphasize normalized neighborhood rank.

Semantic Prototype Stabilization

The method that follows from the benchmark is *Semantic Prototype Stabilization* (SPS). Let x be an EEG epoch, $f_\theta(x)$ the EEG encoder output, and $g(y)$ the frozen GTR embedding of a text target y . Direct inversion asks Vec2Text to decode $f_\theta(x)$ or a diffusion-prior output $h_\phi(f_\theta(x))$ as if it were a valid sentence embedding. SPS instead separates evidence from verbalization. Its assumption is modest: if image-evoked EEG mainly supports a stable object-level variable c , while a generated caption embedding also contains instance

detail and captioner noise ϵ , then direct regression to $g(c, \epsilon)$ asks the EEG model to predict information that may not be reliably present in the signal. SPS has three steps: compute and normalize $z = f_\theta(x)$; retrieve $p^* = \arg \max_{p \in \mathcal{M}} \langle z, p \rangle$ from a declared memory $\mathcal{M}$; and verbalize only the retrieved prototype. The closed memory $\mathcal{M}_{test}$ contains the 200 held-out test prototypes and is used only as a diagnostic rank test. The open memory $\mathcal{M}_{train}$ contains training-class prototypes or raw training captions and contains no held-out test targets. Repeated prototype targets are handled as multi-positive matches during retrieval-style training so that same-class trials are not treated as false negatives.

Results

Ground-Truth Text Embeddings Invert Correctly

Table 1 is a failure diagnostic rather than a reconstruction leaderboard. The ceiling columns show that GTR embeddings can be inverted into text reliably when the correct embedding is given. The brain columns show the opposite: detailed-caption predictions are below the 1,000-derangement real-text null, and label predictions are even more misleading because the fixed template “This picture is” inflates lexical overlap for mismatched labels. The key quantity is therefore the brain-minus-null delta, not the ceiling.

Detailed Captions Are Near Chance from Visual EEG

Despite the high ceiling, detailed caption targets are almost unrecoverable from EEG. The brain suite obtains only 2.33 BLEU and 0.149 token-set F1, below the real-text derangement null’s 4.73 BLEU and 0.222 token-set F1. Caption class means do not solve the problem: they reach 2.46 BLEU and 0.144 token-set F1. This is a useful negative result because it rules out a trivial explanation. Averaging captions within class reduces some instance noise, but the resulting target is still a caption-derived sentence embedding rather than a clean object-concept target.

Coarse Labels Reveal a Decodable Semantic Channel

Class labels are not a solution to sentence reconstruction. Their template makes textual overlap metrics misleading: a mismatched label still obtains 52.59 BLEU and 0.749 token-set F1 because every label begins with the same words. Brain embeddings trained against label targets reach only 12.30 BLEU and 0.349 token-set F1, with exact match 0.0. The useful lesson is not that labels reconstruct well, but that coarse concepts are a better supervision level than detailed captions.

Semantic Prototype Retrieval Is a Better First Metric

Open-ended inversion is slow and sensitive to small embedding errors, so we also evaluate the route proposed by the bottleneck analysis: retrieve a sentence prototype from a semantic target set before verbalization. We first use a closed 200-way diagnostic in which the candidate set is the held-out test target set. This is not a deployable reconstruction

setting, but it directly measures whether EEG embeddings preserve semantic rank. Under uniform random ranking, top-5 chance is 2.5%; the empirical retrieval null in Table 2 is higher because it samples real prototype neighborhoods and is therefore the stricter baseline used for claims. Prototype retrieval is much stronger than free-form sentence inversion: label prototypes reach 20.0% top-5, and caption prototypes reach 38.0–40.3% top-5 in the `sub-08` diagnostic. The diffusion-prior output usually ranks below the direct encoder, supporting the direct-encoder control.

Test-Prototype Retrieval Across Ten Subjects

The main rank diagnostic uses only the 200 held-out test prototypes as candidates. This setting is deliberately closed: it measures whether EEG embeddings preserve semantic rank when the correct target is in the candidate set, but it is not a deployable reconstruction protocol. We therefore also report no-test-target memory results in prose: with the 1,654 training-class prototype memory, the direct encoder reaches 0.426 ± 0.014 retrieved GTR cosine and 0.234 ± 0.009 token-set F1 across ten subjects, compared with $0.307/0.199$ for random retrieval and $0.357 \pm 0.011 / 0.215 \pm 0.003$ for the diffusion prior. THINGS-EEG train and test concepts are disjoint, so this open memory retrieves semantic neighbors rather than exact held-out labels.

Ten-Subject Semantic Prototype Sweep

To test whether the semantic prototype route is specific to `sub-08`, we ran a matched 80-epoch caption-class-prototype sweep on all ten available subjects using the same training and evaluation scripts. Table 2 reports the closed test-200 diagnostic, and Table 3 separately reports the no-test-target memory setting because the two protocols answer different questions. In the closed 200-way diagnostic, the direct encoder reaches $26.25 \pm 5.29\%$ top-5 retrieval across subjects, compared with 4.5% for the retrieval null under the same candidate set. The diffusion-prior output reaches only $9.1 \pm 2.2\%$ top-5, confirming across subjects that the image-generation prior can degrade language-space semantic rank. This encoder-prior gap is significant under paired subject-wise testing: top-5 improves by 0.172 with 95% confidence interval $[0.144, 0.199]$, paired t -test $p = 2.15 \times 10^{-7}$, and Wilcoxon $p = 0.00195$. In the no-test-target train-memory setting, exact top-5 is undefined because held-out targets are absent, but the direct encoder improves retrieved cosine and token F1 over both random and diffusion-prior retrieval. These results convert the route from a single-subject observation into a reproducible ten-subject semantic decoding result, but the absolute retrieval quality remains far from deployable sentence reconstruction.

No-test-target train-bank retrieval, raw-caption memory expansion, dual memory, validation-routed memory, and CSLS are retained as artifact-chain stress tests rather than main results. They are useful for diagnosing memory granularity, but they do not replace the held-out test-200 retrieval setting because the train and test classes are disjoint.

The reproduced MindAlign visual decoder provides a stronger external representation for this same no-test-target protocol. After the exact-code full 150-epoch run, we save

Table 1: Text-inversion diagnostic on subject `sub-08`. The real-text derangement null compares each reference against mismatched held-out text from the same target regime and is the relevant textual chance model. Ceiling values confirm that GTR-to-text inversion works when the correct embedding is known; the negative brain-minus-null deltas show that EEG-derived caption and label embeddings do not support sentence-reconstruction claims.

Target	Null BLEU	Brain BLEU	Ceiling BLEU	Null F1	Brain F1	Δ F1
Detailed captions	4.73	2.33	69.65	0.222	0.149	-0.073
Caption class mean	4.73	2.46	69.65	0.222	0.144	-0.078
Class labels	52.59	12.30	73.81	0.749	0.349	-0.400

Table 2: Closed test-200 SPS rank diagnostic across ten THINGS-EEG2 subjects. The held-out target is in the candidate set, so top-5 is defined; this is a semantic-rank test, not a deployable reconstruction setting. Values are mean $\pm$ population standard deviation.

Route	Top-5	Retrieved cosine	Token F1
Retrieval null	4.5 $\pm$ 0.0	0.259 $\pm$ 0.000	0.223 $\pm$ 0.000
Diffusion prior	9.1 $\pm$ 2.2	0.317 $\pm$ 0.013	0.249 $\pm$ 0.010
Direct GTR-SPS encoder	26.2 $\pm$ 5.3	0.398 $\pm$ 0.020	0.309 $\pm$ 0.023

Table 3: No-test-target train-memory retrieval across ten THINGS-EEG2 subjects. The memory contains only training-class text prototypes or training captions, so exact held-out top-k is undefined; these rows measure semantic-neighbor verbalization quality rather than target reconstruction accuracy.

Route	Retrieved cosine	Token F1
Retrieval null	0.307 $\pm$ 0.000	0.199 $\pm$ 0.000
Diffusion prior	0.357 $\pm$ 0.011	0.215 $\pm$ 0.003
Direct GTR-SPS encoder	0.426 $\pm$ 0.014	0.234 $\pm$ 0.009
Low-rank MindAlign bridge	0.479 $\pm$ 0.005	0.292 $\pm$ 0.008

subject-tagged top checkpoints, encode train and test EEG, and fit only training-set maps from MindAlign EEG embeddings to GTR caption embeddings. The fixed transfer improves class-prototype cosine from 0.426 to 0.465 and raw-caption F1 from 0.254 to 0.283. Selecting ridge alpha on a class-balanced training-validation split and re-fitting on all training pairs raises class-memory cosine to 0.478 and raw-caption F1 to 0.291. A validation-selected low-rank PCA bottleneck gives a marginal further gain to 0.479 class-memory cosine and 0.292 raw-caption F1; its gain over alpha-selected ridge is only about 0.0015, so we treat it as a calibration ablation rather than a new headline method. Route-complementarity, score-fusion, confidence-routing, and candidate-reranker audits remain bounded: oracle selection reaches 0.315 F1, but deployable confidence routing, linear fusion, and learned reranking do not yield a sentence-reconstruction breakthrough. The strongest current visual EEG-to-language evidence is therefore a high-quality visual EEG representation followed by validation-calibrated semantic bridging and constrained no-test-target verbalization.

The MindAlign transfer and CSLS artifacts remain use-

ful for future memory design, but they are not mixed into Table 2 because their candidate banks differ from the test-200 prototype task. All reported confidence intervals and hypothesis tests are paired at the subject level. Unless otherwise stated, intervals are bootstrap intervals over the ten subject-wise paired differences, and top-k values are reported as percentages while cosine and token-overlap metrics are unitless means.

Target Geometry Helps Explain the Failure

Target geometry helps explain the language mismatch. Detailed captions average 14.66 ± 3.43 words and have only 0.582 within-class GTR cosine over the 10 images in each THINGS class, while class labels, object phrases, and canonical object sentences are class-stable with within-class cosine 1.0. The failure is not merely that captions are long: rule-compressed captions average 6.75 words but lower within-class cosine to 0.545 and perform poorly in a compressed-memory smoke. A consolidated granularity ladder shows the same tradeoff under no-test-target retrieval. Raw-caption memory improves token-set F1 over class prototypes by 0.020 but lowers retrieved GTR cosine by 0.059; a dual class+raw memory recovers 0.017 cosine over raw captions while preserving nearly the same token F1. Attribute-token probing gives a more local view of the mismatch: the MindAlign semantic-transfer route recovers material, scene, and action/pose words above a deranged prediction-text null, but the mean gain is small, color is weak, and shape/size words are below null. Visual EEG is therefore asked to learn fine caption variation that is only unevenly supported by the current decoded representation. This is a diagnostic explanation rather than proof that visual EEG contains no instance-level information: encoder capacity, captioner noise, loss weighting, trial averaging, and embedding-domain mismatch remain possible contributors, which is why the benchmark reports ceilings, real-text nulls, direct-encoder controls, and negative calibration ablations together.

Why Image Reconstruction Can Work While Caption Reconstruction Fails

Image reconstruction and sentence reconstruction use different supervision geometry. In EEG-to-image generation, the model can exploit visual feature spaces whose axes are shaped by object appearance, category, texture, and layout. Even if the EEG embedding is coarse, a diffusion model can turn a noisy visual prior into a plausible image. The evalu-

ation often rewards semantic and visual consistency rather than exact instance recovery.

Sentence reconstruction is harsher. A caption embedding encodes not only object category but also attribute choices, relations, and captioner-specific wording. If two images of the same object class receive different captions, their GTR vectors can be far apart even when the EEG response mostly supports the shared object concept. A decoder then verbalizes the wrong neighborhood in language space, producing low BLEU and low token overlap. This explains why class labels are easier: they remove unsupported visual detail and align supervision with the semantic level that visual EEG can plausibly carry.

Visual-to-Semantic Bridge

The analysis above suggests a concrete alternative: preserve the visual nature of the EEG signal, retrieve in a visual foundation-model space, and then attach language to the retrieved visual prototype. We test this route with the cached ViT-H image embeddings used by the image reconstruction pipeline. A train-only ridge map from ViT-H image embeddings to GTR caption embeddings is a poor bridge: on `sub-08`, EEG-predicted visual embeddings mapped through this ridge reach only 6.0% top-5 in the 200-way closed set, and ground-truth visual embeddings mapped through the same ridge reach only 7.0%. This negative result mirrors the GTR ridge calibration result: pointwise cross-space regression can increase paired cosine without preserving useful semantic rank.

Nearest-neighbor visual retrieval is substantially better for the available checkpoints. Using the same `sub-08` image-reconstruction checkpoint, we retrieve the nearest ViT-H prototype and verbalize that prototype with its cached GTR caption. In the closed 200-way diagnostic, this EEG visual-nearest-neighbor route reaches 29.0% top-5, close to the 30.5% top-5 of the directly trained GTR-SPS encoder under the same subject and schedule. Thus visual-nearest-neighbor verbalization is a plausible way to convert visually decoded EEG information into language without forcing EEG into unstable caption embeddings, but it is not reported as a main population result because available visual checkpoints do not cover all ten subjects uniformly.

Semantic Prototype Stabilization

The immediate implication is that the next model should not be trained directly on raw detailed captions. *Semantic Prototype Stabilization* first aligns visual EEG to stable prototypes—class labels, class-mean captions, controlled concept sentences, or retrieved visual exemplars—and generates text only after retrieval. For large candidate memories, SPS can optionally use CSLS to improve exact rank without re-training the EEG encoder.

The training objective should also be retrieval-aware. A standard contrastive term can overemphasize vector direction, while raw MSE constrains both magnitude and direction. The local training script therefore exposes the loss blend explicitly: in generation mode the default encoder objective gives high weight to MSE (0.90 during training and 0.99 for

validation), while retrieval mode uses L2-normalized embeddings for contrastive alignment. In a matched ten-subject loss sweep, retrieval-mode training does not improve the already strong direct-encoder top-5 mean (26.2% for both losses; 6/10 subject wins), but it consistently improves the diffusion-prior output from 9.1% to 13.0% top-5, with 10/10 subject wins and better MRR, retrieved cosine, and token-set F1. Repeated prototypes must be handled explicitly; otherwise different trials with the same class-label embedding become false negatives in the batch. The current evidence suggests that direct encoder embeddings preserve semantic rank better than the diffusion-prior output, while normalized retrieval loss can partially repair the prior’s language-rank degradation. Future reconstruction improvements should therefore select by held-out semantic rank rather than by raw embedding cosine.

This rank-based view also rules out a tempting but incorrect fix. We tested a frozen ridge calibration from training EEG encoder outputs to training GTR targets on `sub-08`. The calibration increases test diagonal cosine from 0.239 to 0.558 at the best tested regularization, but closed top-5 retrieval drops from 30.5% to 12.5%. Thus pointwise embedding regression can make the predicted vector closer to its paired target while still damaging the neighborhood order needed for text retrieval. SPS therefore treats semantic rank and candidate-bank retrieval as the primary evidence, not raw embedding cosine.

SPS should also expose uncertainty before verbalization. A selective-verbalization audit ranks no-test-target raw-caption predictions by their top retrieved similarity and reports coverage-quality tradeoffs without using the reference text to choose thresholds. For the MindAlign raw-score route, verbalizing only the top 50% most confident trials raises token-set F1 from 0.290 to 0.298 across ten subjects; the score-fusion route reaches 0.316 F1 on the top 10% but covers only 20 examples per subject. These gains are bounded, but they support a practical rule: when semantic evidence is weak, the system should abstain or report a retrieved neighborhood rather than force an LLM to hallucinate a sentence.

For visual EEG, the practical route supported by the present evidence is a visual-to-semantic bridge: decode into a visual foundation-model space, retrieve a class or prototype image, then verbalize the stabilized concept with a constrained captioner or text decoder.

For the final auditory sentence reconstruction goal, the supervision should change. Auditory and reading tasks provide time-aligned language stimuli, so the model can learn segment-level mappings from EEG to sentence or phrase embeddings. The visual pilot should therefore be treated as semantic-bottleneck evidence and failure diagnosis, not as evidence that open-ended sentences can be decoded from visual EEG alone. The route suggested by the present audit is *Temporal Semantic Prototype Stabilization*: train a temporal EEG encoder against sentence embeddings, retrieve first from a train-only sentence or phrase memory, compare against real-text derangement, shuffled-EEG, time-shifted, and frequency-prior nulls, and only then allow an LLM to rewrite the retrieved semantic anchors. The artifact chain includes a manifest contract, a mount-ready ridge baseline, a

Table 4: Tail-aligned real ZuCo 2.0 sentence-window route-decision matrix. All rows use the same unique 69-sentence closed candidate bank unless marked as a null. Tail alignment corrects a systematic task-window count offset and yields a weak rank-gate pass for ridge, but the margin over the strongest null is too small for sentence-generation claims.

Route	Top-5	MRR	Gate
Ridge window baseline	0.120	0.098	weak pass
PLS latent baseline	0.111	0.089	fail
EEG kNN baseline	0.108	0.091	fail
Subject-invariant raw EEGNet	0.084	0.080	fail
Raw EEGNet consensus	0.087	0.070	diagnostic only
Mean text prior null	0.082	0.074	null
Time-shifted EEG null	0.111	0.092	null
Real-text derangement null	0.064	0.052	null

source-backed acquisition plan for public reading/listening EEG datasets, a small ZuCo 2.0 task-material text bank with 833 text rows and 747 unique texts, passage-level real ZuCo smoke tests, and trigger-windowed sentence-window pilots.

The sentence-window pilot is the closest artifact to the final task. We cut real ZuCo 2.0 EEG into trigger-10/trigger-11 sentence windows, remove windows shorter than 0.5 seconds, and align windows to task-material sentences. A first head-truncated conversion failed both null gates. Auditing the event counts showed a systematic offset: every EEG file has two fewer sentence windows than task sentences, except NR7 files, which have three fewer. We therefore add an explicit tail-alignment mode that records the sentence offset in the manifest. The resulting dataset has 2,739 rows from 12 subjects with 520 EEG feature columns and 298 unique sentence texts. The split has no exact train-test text overlap and exposes a 69-sentence closed test bank. Under GTR embeddings, the alignment-corrected closed diagnostic reaches 0.026 top-1, 0.120 top-5, and 0.098 MRR. The real-text derangement null has 0.064 top-5 and 0.052 MRR, and time-shifted test EEG reaches 0.111 top-5 and 0.092 MRR. Table 4 consolidates the tail-aligned route-decision matrix. PLS, EEG kNN, and the mean-text prior do not improve over ridge. A subject-invariant raw-window EEGNet improves validation top-5 to 0.167 but generalizes poorly, with test top-5 0.084 and repeated-sentence consensus 0.087. Thus the corrected ingestion path, grouped text splits, unique-candidate nulls, GTR evaluation, baseline matrix, and raw temporal encoder now run on a larger real sentence-window dataset. The result is a weak rank-gate pass, not a sentence-generation breakthrough: the ridge route is above the real-text and time-shifted EEG nulls, but exceeds the strongest null by only 0.0015 top-5. Before any SOTA sentence-reconstruction claim is defensible, this alignment-corrected signal must be strengthened with subject-wise uncertainty, stricter alignment evidence, and a stronger cross-subject temporal objective.

Reproducibility

All results are produced by scripts under EEG_Image_decode/Generation. The vi-

sual EEG-to-language benchmark is driven by the `brain_to_text_benchmark.sh` path and companion audit scripts, while the sentence-EEG extension is driven by `run_sentence_eeg_tsps.sh`. The artifact manifest lists the benchmark, real-text null, ten-subject retrieval sweep, MindAlign transfer audit, loss/adaptor diagnostics, SOTA protocol audit, real ZuCo smoke/pilot artifacts, and claim-support audit. The experiment log records commands and output CSV files, and a verifier checks citations, required result files, and scripts.

Limitations

The results establish a visual EEG-to-language bottleneck analysis and prototype-verbalization route, not arbitrary sentence decoding from visual EEG. The data limitation is that THINGS-EEG2 is a visual object-perception task: its text targets are imposed by captioners or folder labels rather than time-aligned language stimuli. The target limitation is that captioner priors, template wording, and GTR geometry can contaminate text-overlap metrics, even when real-text derangement nulls are used. The model limitation is that some sub-08 target-mode comparisons use one to three seeds, and the ten-subject semantic sweep uses a matched single-seed 80-epoch schedule; longer matched schedules would refine subject-level estimates. The evaluation limitation is that the 200-way prototype retrieval experiment is a closed diagnostic; no-test-target training-memory retrieval is more realistic but retrieves semantic neighbors because train and test concepts are disjoint. The sentence-EEG limitation is that the real ZuCo artifacts currently validate data ingestion and null construction only; they do not establish sentence reconstruction performance. Finally, the generalization limitation is that success on THINGS-EEG2 prototype retrieval should not be interpreted as evidence that auditory sentence reconstruction will work without time-aligned language EEG. Auditory or reading EEG remains the proper setting for open sentence reconstruction.

Ethical Impact

This work should be described as semantic neural decoding, not mind reading. The experiments reconstruct text associated with externally presented stimuli under a controlled offline benchmark. They do not infer arbitrary private thoughts. The long-term application to auditory brain-language interfaces raises privacy concerns, so future systems should report chance controls, ceiling controls, subject-wise generalization, and failure modes before making user-facing claims.

Conclusion

The failure of visual EEG-to-caption reconstruction is informative. Ground-truth GTR embeddings invert well, but EEG-derived embeddings trained on detailed captions fall below a real-text derangement null under free-form text metrics. Test-200 semantic prototype retrieval is above the retrieval null across ten subjects, and no-test-target training-memory retrieval recovers semantic neighbors without using held-out labels as candidates. The bottleneck is therefore semantic target mismatch: visual EEG supports visually grounded

semantic neighborhoods more reliably than exact instance-level sentence detail. A defensible path toward brain-to-text is to first decode stable semantic prototypes, then verbalize them, and to reserve open-ended sentence reconstruction for auditory or reading paradigms with time-aligned language supervision.

References

- Chen, Z.; Liu, S.; Lu, R.; Qi, H.; Woolgar, A.; Wang, X. V.; and Wang, L. 2026. MindAlign: Bridging EEG, Vision, and Language for Zero-Shot Visual Decoding. arXiv:2605.24523.
- Collautti, E.; Mao, X.; Tonin, L.; Tortora, S.; and Puthusserypady, S. 2026. RAG-based EEG-to-Text Translation Using Deep Learning and LLMs. arXiv:2605.17503.
- Grootswagers, T.; Zhou, I.; Robinson, A. K.; Hebart, M. N.; and Carlson, T. A. 2022. Human EEG Recordings for 1,854 Concepts Presented in Rapid Serial Visual Presentation Streams. *Scientific Data*, 9(3).
- Jeon, M.; and Kim, H. 2025. DELTA: Language Diffusion-based EEG-to-Text Architecture. arXiv:2511.21746.
- Jo, H.; Yang, Y.; Han, J.; Duan, Y.; Xiong, H.; and Lee, W. H. 2024. Are EEG-to-Text Models Working? arXiv:2405.06459.
- Li, D.; Wei, C.; Li, S.; Zou, J.; Qin, H.; and Liu, Q. 2024a. Visual Decoding and Reconstruction via EEG Embeddings with Guided Diffusion. arXiv:2403.07721.
- Li, J.; Chen, J.; Shen, F.; Yang, J.; Li, J.; Yuan, Z.; Cheng, B.; Li, M.; and Yang, Y. 2026. Assembling the Mind’s Mosaic: Towards EEG Semantic Intent Decoding. In *International Conference on Learning Representations*.
- Li, J.; Song, Z.; Wang, J.; Zhang, M.; and Zhang, Z. 2024b. BrainECHO: Semantic Brain Signal Decoding through Vector-Quantized Spectrogram Reconstruction for Whisper-Enhanced Text Generation. arXiv:2410.14971.
- Liu, X.; Shen, D.; and Liu, X. 2025. Learning Interpretable Representations Leads to Semantically Faithful EEG-to-Text Generation. arXiv:2505.17099.
- Mishra, A.; Shukla, S.; Torres, J.; Gwizdka, J.; and Roychowdhury, S. 2025. Thought2Text: Text Generation from EEG Signal using Large Language Models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, 3747–3759.
- Morris, J. X.; Kuleshov, V.; Shmatikov, V.; and Rush, A. M. 2023. Text Embeddings Reveal (Almost) As Much As Text. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.
- Murhekar, A.; Liu, C.; Mishra, A.; Roychowdhury, S.; and Gwizdka, J. 2026. SENSE: Efficient EEG-to-Text via Privacy-Preserving Semantic Retrieval. arXiv:2603.17109.
- Ni, J.; Qu, C.; Lu, J.; Dai, Z.; Abrego, G. H.; Ma, J.; Zhao, V. Y.; Luan, Y.; Hall, K. B.; Chang, M.-W.; and Yang, Y. 2021. Large Dual Encoders Are Generalizable Retrievers. arXiv:2112.07899.
- Tang, J.; LeBel, A.; Jain, S.; and Huth, A. G. 2023. Semantic Reconstruction of Continuous Language from Non-Invasive Brain Recordings. *Nature Neuroscience*, 26: 858–866.
- Tao, Y.; Liang, Y.; Wang, L.; Li, Y.; Yang, Q.; and Zhang, H. 2024. SEE: Semantically Aligned EEG-to-Text Translation. arXiv:2409.16312.
- Wang, J.; Song, Z.; Ma, Z.; Qiu, X.; Zhang, M.; and Zhang, Z. 2024. Enhancing EEG-to-Text Decoding through Transferable Representations from Pre-trained Contrastive EEG-Text Masked Autoencoder. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- Wang, Z.; and Ji, H. 2022. Open Vocabulary Electroencephalography-to-Text Decoding and Zero-Shot Sentiment Classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, 5350–5358.
- Yang, X.; Tian, H.; Li, Y.; Zhang, J.; Li, S.; and Pan, G. 2026. Brain-CLIPLM: Decoding Compressed Semantic Representations in EEG for Language Reconstruction. arXiv:2604.16370.
- Zhou, J.; Duan, Y.; Chang, F.; Do, T.; Wang, Y.-K.; and Lin, C.-T. 2024. BELT-2: Bootstrapping EEG-to-Language Representation Alignment for Multi-task Brain Decoding. arXiv:2409.00121.